Final Year Project

Zeroth Review

Group Number :22

Guide : Kunwar Singh

Members

Anurag 106119117

Manmohan 106119072

Sobhagya 106119117

Title

Cryptanalysis of lattice based cryptography

Lattices are mathematical structures that can be used to construct cryptographic schemes such as fully homomorphic encryption, identity-based encryption, and lattice-based signatures.

Scope

The project is to analyze lattice-based cryptography to identify and improve design issues related to efficiency. It will study lattice-based encryption schemes, identify vulnerabilities, and optimize existing algorithms. The goal is to enhance security and efficiency of latticebased cryptography for practical use.

Scope

- The aim of this project is to examine the design and efficiency of lattice-based encryption methods. This will involve evaluating various lattice-based algorithms and pinpointing any weaknesses or vulnerabilities. The project will also focus on finding ways to optimize the performance of these encryption schemes.
- The ultimate objective is to enhance the security and practicality of lattice-based cryptography by addressing any issues with its design and efficiency.

References

- Deng, X. (2016). An Introduction to Lenstra-LenstraLovasz Lattice Basis Reduction Algorithm.
- MATZOV. (2022). Report on the Security of LWE: Improved Dual Lattice Attack. Zenodo. https://doi.org/10.5281/zenodo.6412487
- Mariano, Artur & Laarhoven, Thijs & Correia, Fabio & Rodrigues, Manuel & Falcao, Gabriel. (2017). A Practical View of the State-of-the-Art of Lattice-Based Cryptanalysis. IEEE Access. PP. 1-1. 10.1109/ACCESS.2017.2748179.